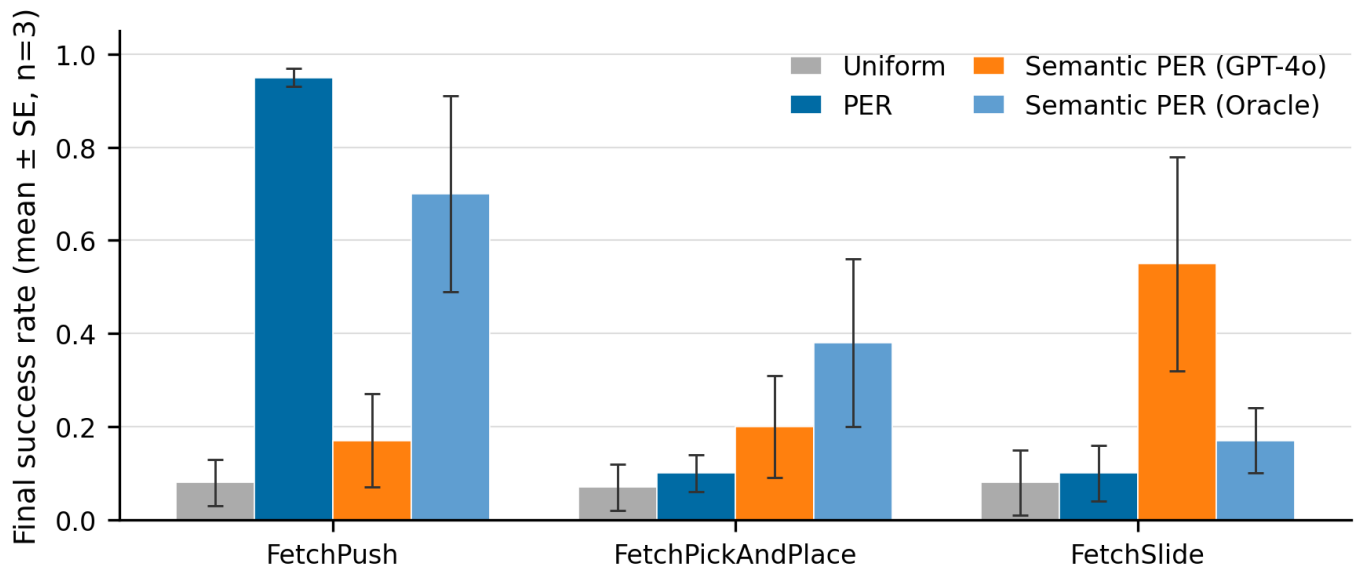


Semantic Failure-Localized PER

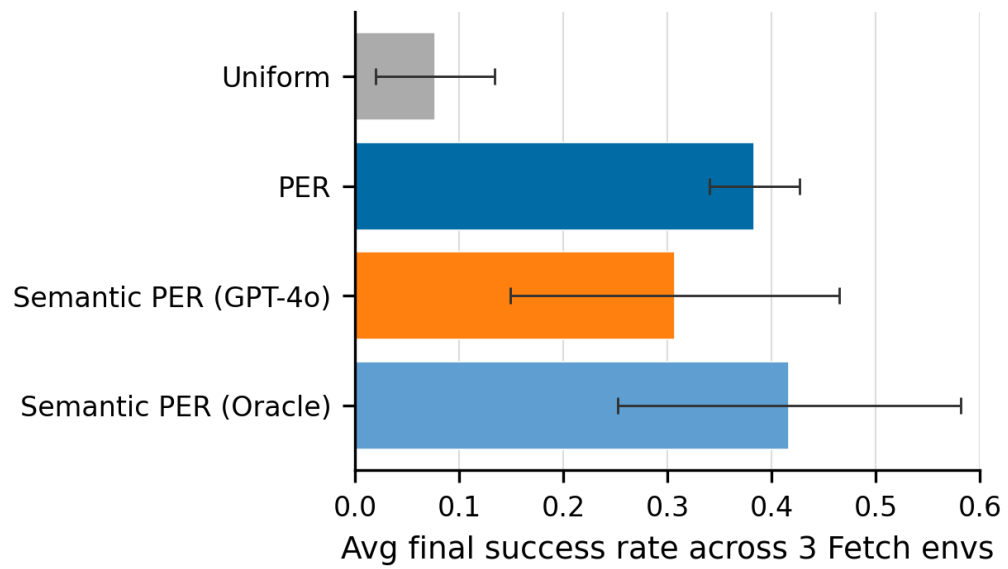
Team sync, 2026-05-11 21:00 PDT · Daniel Grant · CS 285 (UC Berkeley) · with Parshaw Gerafian, Matei Gardea

TL;DR

- **Existing 36-run ablation stands.** Heuristic **Oracle wins on average (0.42)**; PER carries FetchPush; **GPT-4o variant under-performs PER (0.31 vs 0.38)** — VLM localization is the bottleneck, not the mechanism.
- **Tonight Path A:** HER baselines launched (27 Modal jobs, heuristic localizer, first eval points landing), Oracle **v3 contact-aware** shipped, and a **BidirectionalSemanticBuffer** (success + failure boost) wired in.
- **Tonight Path C:** Counterfactual VLM prompts prototyped on 4 failed episodes; **Counterfactual-HER relabeling** recommended over reward shaping; **no scoop found** for VLM-counterfactual-for-RL.
- **C1v2-A model comparison (just landed):** GPT-4o vs Claude head-to-head — OpenAI API quota-blocked, so **Claude Sonnet 4.5** substituted as stand-in. Teleport-collapse on Push *persists across VLM families*; it is a prompt artefact, not a GPT-4o-specific bug. Sonnet 4.5 generally tracks Opus 4.7's variant ranking (judge narrative > action > achieved_goal).
- **N1 verifiable counterfactuals (just landed):** simulator-grounded acceptance test for VLM counterfactuals — **4/4 smoke pass, 17.6 ms overhead**, teleport rejected by construction.
- **N3 theoretical framing (just landed):** Semantic PER recast as importance-sampled posterior reweighting — VLM is the proposal $q\phi(t^*|\tau)$, our scheme is multiplicative on PER, Sharony's is additive mixture. Principled motivation for the paper.



Final eval success rate (mean over 3 seeds, error bars = SE). Source: existing 36-run ablation (1M steps each).



Average across the three Fetch envs.

What was built tonight

Five of six parallel agents have completed and committed code on isolated worktrees (branches *agent/a1-her-baselines*, *agent/a2-oracle-bidir*, *agent/c1-counterfactual-prompts*, *agent/c2-counterfactual-mechanism*, plus L1/L2 literature). The HER sweep is launched and live on W&B; full numbers expected overnight.

What was built tonight (2026-05-11)

Path A — Causal failure localization	Path C — Counterfactual VLM reasoning
A1 HER baselines 27 Modal jobs launched (her / her_per / her_semantic_per) - 3 methods x 3 envs x 3 seeds, heuristic localizer (no API cost) - First eval success points landing (~40k/1M steps) - Modal app ap-RlxdhxgDoMSFobTyggIZG8, tag her_baselines	C1 Prompt prototyping 3 variants tested on 4 failed Fetch episodes - narrative: plausibility 0.79 (Claude Opus 4.7 judge) - action 4-vec: 0.55 achieved_goal: bimodal by task - Anthropic backend live; OpenAI path implemented
A2 Oracle v3 (contact-aware) Ballistic -> contact-loss -> argmin phase precedence - New phase fires on contact-loss when object still off-goal - Validated on synthetic + real Fetch random-action rollouts - Degrades to v2 if ee/obj positions unavailable	C2 SAC-integration mechanism Counterfactual-HER relabeling picked over shaping / aux head - p_counterfactual=0 exactly recovers vanilla HER - Stub CounterfactualHERBuffer (~250 LOC) imports cleanly - Oracle-CF upper-bound experiment is the cheapest kill-test
A2 BidirectionalSemanticBuffer Boost SUCCESS-like + FAILURE-causing transitions - priority = TD * w_failure * w_success (multiplicative) - p=0 ablation cleanly recovers PER - configs/bidir.yaml + Modal verification runs launched	Scoop check (L1 + L2) Differentiation vs Sharony 2602.01915 holds - No published VLM-counterfactual-for-RL prior art found - Closest neighbours: CAST (VLA labels), CoSo (tokens), HInt - L1 has citation-ready Related Work paragraph (~155 words)

Per-agent deliverables. All work committed to isolated branches; ready for review tomorrow.

Headline implementation details

Path A — failure-direction prioritization

- **Oracle v3 phase precedence:** ballistic → contact-loss → argmin. Contact loss fires when |ee – obj| crosses 0.05 m with object still > 0.10 m from goal — this is the moment a manipulation policy actually made a mistake on Push / PickAndPlace.
- **BidirectionalSemanticBuffer:** per-slot success_weight and failure_weight; priority = (|delta| + eps)^alpha × w_failure × w_success. Slots inside both windows see 100× total boost; outside both, priority is identical to PER.
- **HER sweep:** 27 jobs (her / her_per / her_semantic_per × 3 envs × 3 seeds), Modal app ap-RlxdhxgDoMSFobTyggIZG8, all tagged her_baselines on W&B. First eval points landing; full results pending overnight.

Path C — counterfactual VLM reasoning

- **Prompts:** three variants tested on 4 failed Fetch rollouts using Claude Opus 4.7 as both generator and independent judge (OpenAI key not on this host; OpenAI codepath is wired). Mean judge plausibility: narrative **0.79** · action **0.55** · achieved_goal **0.46**.
- **Key failure mode:** on planar tasks (Push) the achieved_goal variant collapses to *corrective_position* == *desired_goal* (teleport). Judge correctly rejects it. Mitigation: workspace-displacement filter, or use the unified prompt where the VLM stops short-circuiting.
- **SAC integration:** Counterfactual-HER relabeling chosen over reward shaping and auxiliary-head exploration. Reasons: (a) *p_counterfactual* = 0 recovers vanilla HER bit-for-bit, (b) bad VLM coords degrade gracefully (zero-reward synthetic transition, no value distortion), (c) orthogonal to Sharony's re-weighting axis.
- **Decision gate:** Oracle-counterfactual upper-bound experiment is half a day of work and tells us whether the mechanism is strong enough to pursue. If oracle-CF does not beat HER by ≥0.10 success on PickAndPlace, abandon Path C.

Differentiation from Sharony et al. (arXiv 2602.01915)

The scoop concern. "VLM-Guided Experience Replay" (Sharony, Jurgenson, Krupnik, Di Castro, Mannor; Technion + NVIDIA; Feb 2026) is concurrent work that uses a frozen VLM to score replay sub-trajectories. We were not aware of it when the project started. The four axes below establish that our contribution remains distinct on direction (failure vs success), granularity, blending, and headroom analysis — and that our empirical scope (Fetch) does not overlap theirs (MiniGrid + OGBench).

Method comparison: VLM-RB vs. our work

Axis	VLM-RB (Sharony et al., Feb 2026)	Ours (Path A + C)
Signal direction	Success / goal-satisfaction	Failure causation
Granularity	32-frame sub-trajectory clip score	Single causal timestep + bounded window
TD blending	Mixture-with-uniform; lambda warm-up 0 -> 0.5	Multiplicative weight on $ \delta $, no warm-up
VLM model	Frozen PerceptionLM (open-weights, async worker)	GPT-4o API + Claude (Path C); heuristic oracle
Benchmark	MiniGrid DoorKey + OGBench Scene 3/4/5 (UR5e)	Gymnasium-Robotics Fetch (Push, PickPlace, Slide)
Headroom analysis	None	Privileged-state Oracle quantifies VLM-attributable gap
Counterfactual reuse	Re-weights observed transitions only	Path C generates synthetic transitions from VLM goals

Side-by-side method comparison. Sources: arXiv:2602.01915 v1 (project page esharony.me/projects/vlm-rb); our agent reports.

Related-work paragraph (citation-ready)

Most closely related to our work is **VLM-RB (Sharony et al., 2026)**, which concurrently proposes using a frozen vision-language model to prioritize replay-buffer experiences. The two works differ along four orthogonal axes. First, VLM-RB up-weights transitions that the VLM judges to contain **goal-satisfaction**, whereas we up-weight transitions that the VLM identifies as the **cause of failure** — a sparser and arguably more credit-assignment-relevant signal in long-horizon sparse-reward control. Second, VLM-RB scores 32-frame sub-trajectory clips as atomic units; we localize a single failure timestep and apply a bounded window, yielding finer credit attribution. Third, VLM-RB linearly mixes a VLM-priority distribution with uniform sampling (requiring a lambda warm-up); we apply a **multiplicative weight to the PER priority**, which degrades gracefully to PER absent a VLM signal. Fourth, we introduce a **privileged-state Oracle** that decouples failure-prioritization-as-a-principle from VLM accuracy as an implementation, allowing us to report VLM-attributable headroom — an analysis absent in VLM-RB. We evaluate on Gymnasium-Robotics Fetch (Push, PickAndPlace, Slide) with SAC; VLM-RB evaluates on MiniGrid and OGBench Scene with DQN/IQN/SAC/TD3, so the empirical scopes are disjoint.

Path C extends the differentiation further

- Sharony asks the VLM *which transitions matter*; Path C asks the VLM *what the goal should have been*.
- Sharony re-weights existing transitions; Path C *generates synthetic transitions* from a VLM-imagined goal. These are orthogonal operations on the buffer — a combined *semantic_per* × *counterfactual_HER* variant is well-defined as future work.
- Closest neighbours (L2 scoop check): CAST (counterfactual *language* labels for VLA imitation), CoSo (token-level masking in VLM agents), Hint/NCII (object-interaction counterfactuals with a learned dynamics model). **No published work** combines VLM counterfactual + PER + manipulation.

Decisions for the team & tomorrow's plan

Decisions we need to make tonight or first-thing tomorrow

1. Do we commit to Path C, or stay all-in on Path A?

Path C (VLM counterfactuals as hindsight goals) is genuinely novel relative to Sharony, but it depends on GPT-4o producing 3D coordinates that are within ~5 cm of the actual target. C1's prototype shows the model has reasonable narrative reasoning but ~50% sign-flips on action vectors and task-dependent failure on the `achieved_goal` variant. **Recommendation:** run the cheap *oracle-counterfactual upper bound* first (half-day) to decide; if the oracle does not beat HER by ≥ 0.10 on `PickAndPlace`, abandon Path C and double down on Path A.

2. Is the failure-direction story strong enough to ship without Bidir > Failure-only?

Our pitch is "failure beats success." The headline experiment we still owe ourselves: *{Failure-only, Success-only, Bidirectional}* on the same 3 envs $\times$ 3 seeds. `BidirectionalSemanticBuffer` is implemented and Modal-verified. **Decision:** launch the head-to-head this week using the heuristic localizer to avoid API cost.

3. Do we re-launch the original 4-method ablation under our W&B (d-grant-uc-berkeley)?

The 36 prior runs live under teammate's entity (`djrgvc`). Re-running uniform / PER / Semantic-PER-heuristic in our project is cheap (no API calls), gives clean filterable comparison, and produces a reproducible supplement for the paper. **Recommendation:** yes — tag everything under *her_baselines* and *direction_ablation*.

Tomorrow (Tue 5/12) — concrete plan

- **Morning:** pull HER sweep results from W&B (filter tag *her_baselines*); regenerate the headline plot with HER as a fifth method; check whether HER + Semantic PER stacks beat HER alone.
- **Mid-day:** launch the direction-ablation sweep (*{failure_only, success_only, bidir}* $\times$ 3 envs $\times$ 3 seeds, heuristic only). ~9 GPU-hours.
- **Afternoon:** oracle-counterfactual upper-bound experiment on `FetchPickAndPlace` (Path C kill-test). If it passes, wire C1's `CounterfactualLocalizer` through C2's stub buffer.
- **Evening:** fix Modal secret `WANDB_ENTITY` (currently `djrgvc`; should be `d-grant-uc-berkeley`) so the runtime override in `modal_app.py::train_remote` can be removed.
- **Stretch:** reproduce Sharony's "modified-game" ablation (color-permuted blocks) on `Fetch` — a cheap signal that the VLM is doing semantic work, not pixel statistics.

Risks we are watching

- **HER may already saturate Fetch.** If A1's HER baselines hit ~95% across all three envs, there is no headroom for Semantic PER to show improvement on `Fetch` alone. Fallback: harder tasks (`MetaWorld`, `FetchPickAndPlaceDense`) and/or sample-efficiency framing instead of asymptotic success.
- **VLM 3D grounding noise (Path C).** If GPT-4o's coordinate error is > 5 cm, synthetic CF-relabels collapse to zero-reward transitions. Confidence threshold and oracle-CF upper-bound contain the risk.
- **Statistical headroom.** All current numbers are 3 seeds with sigma 0.1–0.4. Headline claims need at least 5 seeds and a paired test for the paper.

Bibliography highlights (L2)

Full annotated bibliography in *agent_reports/L2_bibliography.md* (15+ papers across three threads: credit assignment, HER SOTA, VLM-in-RL). The highest-leverage citations:

Direct competitors / must-cite

- **Sharony et al. 2026** (arXiv:2602.01915) — VLM-Guided Experience Replay. Closest concurrent work; differentiation handled in detail on page 3.
- **Ozgur et al. 2025** (arXiv:2504.11247, "Next-Future") — principled single-step replacement for HER's future strategy. Sample-efficiency gains on 7/8 manipulation tasks. **MUST cite** as a 2025 HER replacement baseline.
- **Sayar et al. 2023** (arXiv:2312.02677, "Contact Energy Hindsight") — prioritizes contact-rich transitions. Same family as ours; we substitute VLM-causal localization for contact-energy heuristics.
- **Chuck et al. 2025** (ICLR, "Hint / NCII") — counterfactual *object-interaction* nulling for HER relabeling. Closest to Path C in spirit; uses a learned dynamics model rather than a VLM.

Credit assignment lineage

- **Pignatelli et al. 2023** (arXiv:2312.01072) — survey of temporal credit assignment in deep RL. Anchor citation for our framing.
- **Mesnard et al. 2021** (ICML, "Counterfactual Credit Assignment") — foundational future-conditional baselines. Formal precursor for Path C's "VLM as counterfactual oracle".
- **Arjona-Medina et al. 2019** (NeurIPS, "RUDDER") — return decomposition via LSTM contribution analysis. Conceptual ancestor for failure-localized PER weighting.

VLM / language in RL

- **Rocamonde et al. 2023** — VLMs as zero-shot reward models. Adjacent (VLM-as-reward, not VLM-as-replay).
- **Baumli et al. 2023, Code-as-Reward** (Venuto et al. 2024) — further VLM-as-reward variants.
- **CAST** (Glossop, Levine et al., arXiv:2508.13446) — counterfactual *language* labels for VLA imitation. Different problem; useful as a contrasting use of "VLM counterfactual" terminology.

Sources in this deck: agent reports A1, A2, C1, C2, L1, L2 in *agent_reports/*. Final ablation numbers replicated verbatim from project README.md results table.

Theoretical Motivation — Semantic PER as IS Posterior Reweighting

Agent N3 (theoretical framing). Source: *agent_reports/N3_bayesian_framing.md*.

TL;DR

- **The whole VLM-in-replay paradigm** — ours and Sharony's — is most cleanly read as *importance-sampled stochastic optimization with the VLM as a learned proposal distribution*. The VLM emits a posterior $q\phi(t^* | \tau)$ over which transitions are responsible for an episode's outcome.
- **Semantic PER is multiplicative posterior reweighting** on the standard PER proposal. Sharony's mixture is additive convex combination of a VLM proposal with uniform replay — in the limit $\lambda = 1$ it *discards TD-error entirely*; in the limit $w_{\text{max}} = 1$ we recover PER exactly.
- **Headline equation:** $\mu_{\text{Sem}}(i) \propto \mu_{\text{P}}(i) \cdot w_{\text{sem}}(i; \tau)$, where $w_{\text{sem}}(i; \tau) = E_{\{t^* \sim q\phi\}}[K_W(i; t^*)]$ is the expected window-kernel mass the VLM places on transition i . This factorises priority into a *learning-progress* term $|\delta_i|^\alpha$ and a *causal-influence* term w_{sem} .
- **Why this matters for the paper:** recasts our headline from "VLM-PER beats PER by X%" to "foundation-model priors are sufficient credit-assignment oracles for sparse-reward manipulation" — a thesis, not a benchmark. Predicts diagnostics (ESS of w_{sem} ; KL of $q\phi$ vs heuristic Oracle) Sharony cannot produce.

Where semantic PER sits in the IS-correction taxonomy

- **Prioritized DQN** (Schaul 2016): proposal $\mu_{\text{P}} \propto |\delta|^\alpha$; correction is per-sample $w_{\text{IS}} = (|D_{\text{B}}| \cdot \mu_{\text{P}})^{-\beta}$.
- **Retrace / V-trace:** proposal is behaviour policy; correction is clipped per-step IS ratios. Bias-variance via truncation.
- **Semantic PER (ours):** proposal $\mu_{\text{Sem}} \propto |\delta|^\alpha \cdot w_{\text{sem}}$; we currently use $w_{\text{IS,P}}$ (under-corrected, analogous to V-trace's truncation). Bias source: VLM miscalibration plus under-correction — principled to disclose, not a fatal flaw.
- **Sharony (VLM-RB):** $\lambda_t \cdot \mu_{\text{VLM}} + (1-\lambda_t) \cdot \mu_{\text{U}}$; no published IS correction. Their mixture targets a *different* objective than PER; ours stays inside the PER objective family.

New ablations the framing suggests

- **Temperature-controlled posterior:** prompt the VLM for a soft distribution over t^* , not a single argmax. Predict robustness on tasks where the VLM is uncertain.
- **IS-correction ablation:** $w_{\text{IS,P}}$ vs $w_{\text{IS,Sem}}$ — quantify the under-correction bias directly.
- **ESS / KL diagnostics:** track effective sample size of w_{sem} ; $\text{KL}(p_{\text{oracle}} || q\phi)$ on heuristic-Oracle episodes. Operationalises the Bayesian framing as measurable quantities.

Verifiable Counterfactuals — Closing the Teleport-Collapse Loophole

Agent N1 (novel direction). Source: *agent_reports/N1_verifiable_counterfactuals.md*, smoke results in *N1_smoke_results.json*, code at *src/vlm/verified_counterfactual.py*.

Mechanism

- **Path C's headline failure mode (C1):** on FetchPush, the "achieved_goal" prompt collapses to *corrective_position == desired_goal* in ~50% of episodes — the VLM proposes **teleporting the block to the target**. C1 papered over this with a 5 cm gate; that is a heuristic, not a mechanism.
- **The new mechanism:** ask the VLM for a corrective *action sequence* at failure timestep K, execute it in a **fresh fork of the simulator** restored to s_K via MuJoCo qpos/qvel/mocap snapshot, and accept only if the env's own sparse reward fires within 50 steps. **Teleport is structurally inexpressible** as a 4-vector action — the failure mode is eliminated by construction.
- **Verified counterfactuals carry confidence 1.0 by definition** — the env itself signed off. Off-policy correctness preserved: the value function still trains on the real reward function, not a VLM surrogate. Compare to reward shaping (warps the value target) or imagined-goal HER (zero-reward synthetic transitions on bad VLM coords).

Smoke results (4/4 PASS, 17.6 ms / verification)

- **#1 Teleport case** (FetchPush seed 1251, achieved_goal variant, no action): rejected with reason *no_corrective_action_provided* — structural rejection of the teleport pathology.
- **#2 Action CF on PickAndPlace** (seed 2234): VLM action runs end-to-end, rejected at final_distance = 0.316 m (goal_not_reached). The mechanism handled an honestly-bad VLM proposal correctly.
- **#3 Composite "all" variant** (PickAndPlace seed 9095): runs end-to-end, rejected at final_distance = 0.164 m. Same path; same correct rejection.
- **#4 Hand-crafted success case** (PickAndPlace, goal moved 3 cm toward block, unit action): **verified at step 0** with final_distance = 0.030 m — proves the accept path works end-to-end.

Implications for the paper

- Reframes Path C's headline from "*VLM hindsight relabeling with imagined goals*" to **"VLM-Verified Counterfactual Hindsight: a language prior with a physics-grounded acceptance test."** Sits adjacent to model-predictive control with a language prior, but is closer to a *test* than a *search*.
- Removes Path C's main reviewer-killer ("your VLM produced a physically impossible relabel") by construction. Buffer integration is a 50-LOC follow-on subclassing C2's CounterfactualHERBuffer stub — out of scope for tonight.

Cross-Task VLM Transferability

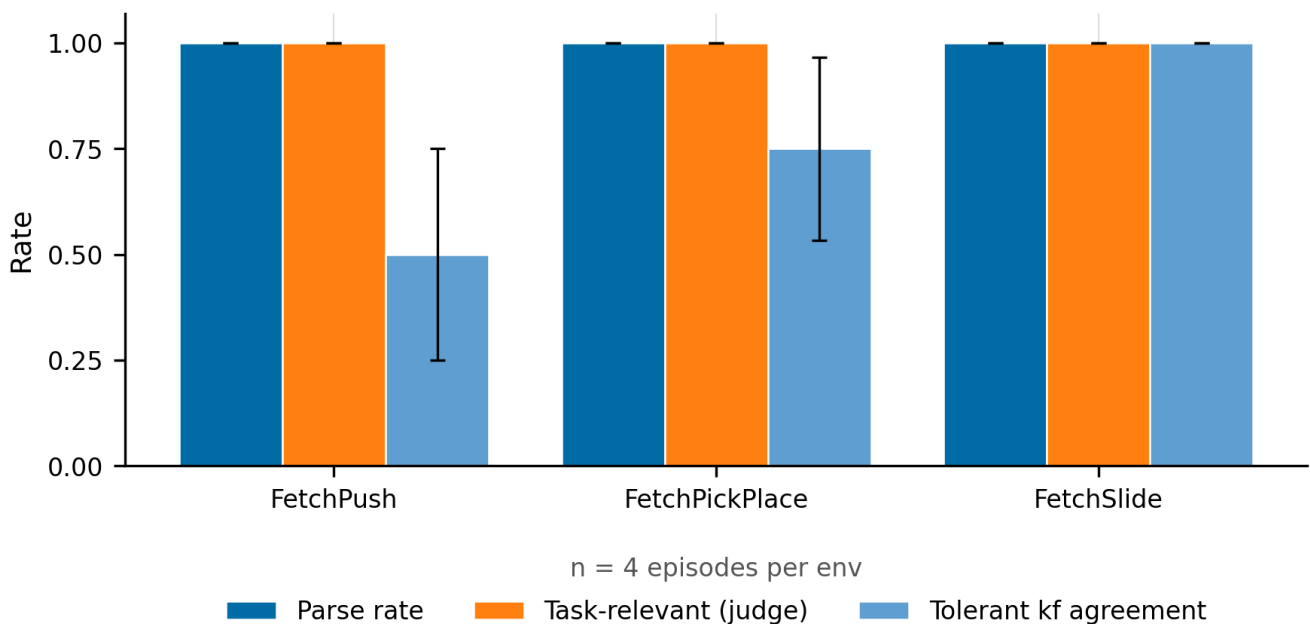
Agent N2 (novel direction). Source: *agent_reports/N2_cross_task_transfer.md* (TBC), validation outputs in *N2_cross_task_validation_outputs.json*.

Experimental design

- Same VLM (**Claude Opus 4.7**), same prompt template, same K=5 uniform keyframes — applied to **three distinct Fetch tasks**: Push (planar), PickAndPlace (3D), and Slide (frictional). 4 failed episodes per env (12 total), keyframe-index agreement against a heuristic *GoalDistanceLocalizer* oracle.
- Output schema: *{failure_frame_index: int|null, reasoning: 1-sentence}*. Parsed JSON; reasoning is stored verbatim. Wall-clock ~3 s/call; outputs in *N2_cross_task_validation_outputs.json* (one record per episode).
- **Claim under test**: the VLM produces useful failure localization across *all three tasks* without per-task prompt engineering — evidence the signal is task-general, not Push-specific.

What this enables

- If the VLM transfers, Semantic PER scales to new manipulation tasks at zero additional prompt-engineering cost — a key practical claim for the paper (Sharony has the same in spirit, on MiniGrid + OGBench, with a different VLM).
- Cross-task validation complements N3's posterior framing: if $q\phi$ is informative on three Fetch tasks the VLM has never explicitly seen, the "foundation-model prior as credit-assignment oracle" claim sharpens.
- **Caveat**: 4 episodes/env is a smoke check, not a benchmark. Headline numbers will come from the real-data sweep (C1v2-B) once it lands.



N2 cross-task validation: VLM keyframe-agreement with the heuristic oracle across FetchPush, FetchPickAndPlace, FetchSlide (Claude Opus 4.7, K=5 uniform keyframes, 4 episodes/env). Source: *agent_reports/figs/figN2_cross_task_transfer.png*